Notes: Weller (RFS, 2018)

Does Algorithmic Trading Reduce Information Acquisition?

Contents

1 Big picture

- **Question.** Does algorithmic trading (AT) increase the information content of prices, or can it actually reduce the amount of information that gets acquired before major firm news?
- **Setting.** The paper studies U.S. common stocks around quarterly earnings announcements from 2012 to 2016. The focus is not on how fast prices respond to already public news, but on whether market participants acquire information *before* that news is publicly disclosed.
- **Core conceptual distinction.** The paper separates *informational efficiency* from *price informativeness*. AT may help prices impound *acquired* information more quickly, yet still discourage the *acquisition* of new information.
- **Main challenge.** Information acquisition is not directly observable. One therefore needs a measure of how much information reached prices before a disclosure, relative to how much information was potentially available to acquire.
- **Main measurement contribution.** The paper develops the *price jump ratio*, a stock-quarter event measure that compares announcement-period price jumps with total announcement-related abnormal return variation.
- **Main data contribution.** The paper combines that measure with SEC MIDAS data, which permit construction of several order-book-based proxies for algorithmic trading at the stock-day level.
- **Main empirical result.** More AT is associated with *less* pre-announcement information acquisition. Using lagged log stock price as an instrument for AT, a one-standard-deviation increase in AT reduces information acquisition before earnings announcements by about 9% to 13%.
- **Timing result.** The information gap between high-AT and low-AT stocks opens well before the announcement and keeps widening throughout the pre-announcement month.
- **Robustness result.** The negative relation survives alternative proxies for AT, alternative measures of information in prices, portfolio-based intraperiod timeliness tests, and an extension from earnings announcements to other firm events such as product releases, executive changes, and M&A news.
- **Why this matters.** Much of the AT literature emphasizes millisecond-level price efficiency. This paper shows that improvements in ultra-short-run efficiency can coexist with a deterioration in medium-run price informativeness.
- **Economic takeaway.** The paper's message is not "AT is bad for price discovery" in every sense. It is that technology can improve the *incorporation* margin while worsening the *acquisition* margin, and the second effect may dominate for economically important decisions.

2 Data and measurement (key objects)

2.1 Empirical setting: why earnings announcements are useful

- The main laboratory is quarterly earnings announcements.
- The paper argues that earnings announcements are especially useful because:

- they are among the recurring firm events with the largest average stock-price effects,
  - they are scheduled public events, so informed-trading models make sharp predictions for pre-event price drift,
  - and they occur frequently enough to generate a large panel.
- The paper is explicit, however, that this setting has external-validity limits:
  - firms may simply wait until after earnings announcements to condition decisions on prices, muting welfare effects in this particular setting;
  - and earnings announcements are only one subset of all value-relevant firm news.

Interpretation. Earnings announcements are chosen because they maximize signal-to-noise for measuring pre-disclosure information acquisition, not because they are the only place where AT matters.

2.2 Data sources and sample construction

The paper combines several sources:

- **CRSP** for returns, prices, market capitalization, and return volatility.
- **TAQ** for quoted bid-ask spreads.
- **I/B/E/S** for earnings announcement dates and times and analyst coverage.
- **13F holdings data** for institutional ownership.
- **SEC MIDAS** for stock-day exchange-level order and trade message aggregates used to construct AT proxies.

The main analysis includes all CRSP common stocks matched to I/B/E/S and MIDAS. The sample contains:

- 3,839 unique securities,
- 54,879 stock-quarter earnings-announcement observations,
- and a time span from January 2012 to September 2016.

Announcements recorded after market close are moved to the next trading day, and observations are excluded when the distance from the prior earnings announcement is below 45 days or above 150 days.

2.3 Rolling measurement windows

For each stock-event pair (i, t):

- AT proxies are averaged over the 21 trading days before the announcement, $[T - 21, T - 1]$.
- Controls such as log price, log market capitalization, return volatility, and quoted spreads are measured over an earlier window, $[T - 42, T - 22]$, to ensure predetermination.

Interpretation. The lag structure is important: the paper wants AT and controls to be measured before the information-acquisition window so that they are not mechanically responding to the very price movements being studied.

2.4 The price jump ratio

The paper starts from the cumulative abnormal return

$$CAR_{it}(k_1, k_2) = \sum_{t=k_1}^{k_2} \left(r_{it} - \alpha_i - \sum_{m=1}^{M} \beta_{im} r_{mt} \right), \tag{1}$$

where abnormal returns are computed relative to a Fama–French three-factor model.

The main object is the price jump ratio:

$$\text{jump}_{it}(a, b) = \frac{CAR_{it}(T-1, T+b)}{CAR_{it}(T-a, T+b)}. \tag{2}$$

In the empirical implementation the paper sets $a = 21$ and $b = 2$, so the numerator is the announcement response and the denominator is the total abnormal return variation from 21 trading days before the announcement through two trading days after it.

Interpretation.

- A *high* price jump ratio means most information arrives only when the announcement is released publicly.
- A *low* price jump ratio means prices already drifted substantially toward the post-announcement value before the disclosure.
- In that sense, lower jump ratios mean *more* information acquisition before the event.

2.5 Why the price jump ratio is the right object

The paper argues that the price jump ratio improves on absolute cumulative abnormal returns for two reasons:

- it normalizes by total event-related variation, so it measures information entering prices *relative* to the amount of information that seems available;
- and it is more "model free" than trade-based measures, because it only requires that some market participants move prices in response to information acquired before disclosure.

The author also emphasizes two innovations relative to earlier jump-ratio papers:

- the measure is estimated at the stock-quarter level in panel form,
- and it uses total event-related abnormal return variation as the normalizing benchmark.

2.6 Implementation details and event filter

Abnormal returns are estimated from a 365-calendar-day window ending 90 days before the earnings announcement, and observations with fewer than 63 valid prior trading days are excluded.

Because the price jump ratio is unstable when its denominator is near zero, the paper keeps only economically important events satisfying

$$|CAR_{it}(T-21, T+2)| > \sqrt{24}\,\hat{\sigma}_{it}, \tag{3}$$

where $\hat{\sigma}_{it}$ is daily return volatility measured before the event.

- This is a strict filter.
- About 45.5% of observations survive.
- The paper argues that these are exactly the observations where the information-content question is most meaningful.

Interpretation. Near-zero events have low signal-to-noise and little scope for meaningful information gaps, so dropping them is both statistically and economically sensible.

2.7 Algorithmic trading proxies

Using MIDAS, the paper constructs four AT proxies:

- **Odd-lot volume ratio:** odd-lot volume divided by total volume.
- **Trade-to-order volume ratio:** traded volume divided by order volume.
- **Cancel-to-trade ratio:** number of cancellations divided by number of trades.
- **Average trade size:** traded shares divided by number of trades.

The expected sign convention is:

- higher odd-lot and cancel-to-trade ratios indicate *more* AT,
- higher trade-to-order ratios and larger average trade sizes indicate *less* AT.

Other measurement details:

- the proxies are logged after averaging over $[T-21, T-1]$,
- extreme values below the 1st percentile or above the 99th percentile are dropped,
- and NYSE/NYSE MKT are excluded from some measures because their message structure is not comparable to the order-based feeds of the other exchanges.

2.8 Controls

The main controls are:

- log market capitalization,
- log share price,
- log return volatility,
- quoted bid-ask spread,

- log analyst coverage,
- institutional ownership ratio.

Interpretation. These controls are designed to absorb the most obvious determinants of both AT intensity and pre-announcement information production.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual setup: two components of price discovery

The paper's conceptual point is that price discovery has two pieces:

- **information acquisition:** gathering new value-relevant information before it becomes public,
- **information incorporation:** impounding already acquired information into prices.

Interpretation. Much prior AT research focuses on the second piece. Weller's contribution is to ask whether improvements in the second piece reduce incentives for the first.

3.2 Mechanisms linking AT to information acquisition

The net effect of AT is theoretically ambiguous.

- **Positive channels.**
  - Lower transaction costs and smarter execution can raise the trading profits from an informational edge.
  - Fast liquidity takers or algorithmic analysts may process public signals more effectively.
- **Negative channels.**
  - Algorithmic liquidity providers may better screen informed order flow and avoid adverse selection, reducing information rents.
  - Order-anticipation or "piggybacking" strategies may erode the profits of slow information acquirers.

Interpretation. The paper is not assuming AT must be harmful. The empirical question is which force dominates in equilibrium.

3.3 Baseline panel specification

The main reduced-form specification is

$$\text{jump}_{it}^{(21)} = \alpha + \beta x_{it} + \gamma \, \text{controls}_{it} + \varepsilon_{it}, \tag{4}$$

where x_{it} is one of the four AT proxies.

Interpretation of signs.

- For proxies that *increase* with AT (odd-lot, cancel-to-trade), a positive β means more AT $\Rightarrow$ a higher jump ratio $\Rightarrow$ less pre-announcement information acquisition.
- For proxies that *decrease* with AT (trade-to-order, average trade size), a negative β has the same interpretation.

The paper estimates:

- specifications with only market capitalization,
- richer specifications with the full control set and month fixed effects,
- and specifications with stock fixed effects.

3.4 Main OLS magnitudes

The paper finds strong and consistent baseline effects. For the first specification:

- a one-unit increase in log odd-lot ratio raises the jump ratio by 0.074,
- which corresponds to about a 13% decline in the pre-announcement share of earnings-announcement price impact for a one-standard-deviation increase in odd-lot activity.

Across proxies, the baseline implied declines in information acquisition are roughly:

- 13% using odd-lot ratios,
- 10% using trade-to-order ratios,
- 9% using cancel-to-trade ratios,
- 15% using average trade size.

Interpretation. The result is not hanging on one proxy. All four measures tell the same story.

3.5 Why OLS may be biased

The paper is very explicit that OLS need not be causal. Bias can arise because:

- AT may sort into stocks with less information acquisition,
- AT may withdraw when informed traders are present,
- omitted liquidity conditions may jointly affect AT and the jump ratio.

Interpretation. Reverse causality is a real issue: lower observed AT could be the result of more informed trading, not its cause.

3.6 Instrumental-variables design

To address endogeneity, the paper estimates:

$$x_{it} = \zeta + \eta \, \mathrm{lprice}_{it} + \theta \, \mathrm{controls}_{it} + \delta_{it}, \tag{5}$$

$$\mathrm{jump}_{it}^{(21)} = \alpha + \beta \hat{x}_{it} + \gamma \, \mathrm{controls}_{it} + \varepsilon_{it}. \tag{6}$$

The instrument is the lagged log stock price, measured as the log of the average price over $[T-42, T-22]$.

Economic logic of the instrument.

- Because of the one-cent minimum tick size under Reg NMS, high-priced stocks have a finer price grid in basis-point terms.
- That finer grid increases the need for rapid quote updates and stale-quote monitoring.
- Hence higher nominal stock prices make algorithmic trading relatively more attractive.

3.7 Why lagged price may satisfy the exclusion restriction

The paper argues that, conditional on controls such as market capitalization, spreads, and institutional ownership, the component of nominal share price orthogonal to those variables should have no obvious direct effect on incentives to acquire information.

The discussion addresses:

- liquidity or clientele explanations for target price ranges,
- stock-split signaling stories,
- and the possibility that the instrument is contaminated by nominal price management.

The paper also shows robustness to:

- dropping observations near stock splits,
- and imposing minimum price thresholds such as \$1, \$20, or \$50.

3.8 IV results

The IV estimates reinforce the OLS results. For example, the coefficient on the odd-lot proxy is about:

- 0.0605 in the simple IV specification,
- 0.0507 with the fuller control set.

For the other proxies, the IV coefficients also all have the expected signs:

- trade-to-order ratio: around -0.0920 and -0.0778,
- cancel-to-trade ratio: around 0.104 and 0.0874,
- average trade size: around -0.123 and -0.108.

Kleibergen–Paap statistics are very large in every specification, so weak-instrument concerns are not first order in the paper's implementation.

Interpretation. Once the endogenous response of AT to informed trading is addressed, the negative effect of AT on information acquisition remains strong.

3.9 Timing analysis: price response ratios

To study *when* information acquisition is deterred, the paper defines the price response ratio:

$$\text{responseratio}_{it}^{(k,21)} = \frac{CAR_{it}(T-21, T-k)}{CAR_{it}(T-21, T+2)}. \tag{7}$$

Then for each $k = 1, \ldots, 21$ it estimates

$$x_{it}^{(k)} = \zeta + \eta \, \text{lprice}_{it}^{(k)} + \theta \, \text{controls}_{it}^{(k)} + \delta_{it}, \tag{8}$$

$$\text{responseratio}_{it}^{(k,21)} = \alpha + \beta^{(k)} \hat{x}_{it}^{(k)} + \gamma \, \text{controls}_{it}^{(k)} + \varepsilon_{it}. \tag{9}$$

Interpretation.

- This traces out the information gap day by day through the month before the announcement.
- If AT deters information acquisition continuously, the response-ratio coefficients should become increasingly negative as the disclosure date approaches.

3.10 Alternative measure 1: price nonsynchronicity

The paper also uses a stock-level price nonsynchronicity measure based on

$$r_{it} = \alpha + \beta_i^m r_{mt} + \gamma r_{\text{ind},t} + \varepsilon_{it}, \tag{10}$$

with price nonsynchronicity defined as $1 - R^2$.

The key problem is that AT may also make prices more synchronized through statistical arbitrage, which would mechanically lower $1 - R^2$ even if firm-specific information acquisition did not change.

To deal with this, the paper studies the *difference* between event-period and normal-period nonsynchronicity:

$$x_{it} = \zeta + \eta \, \text{lprice}_{it} + \theta \, \text{controls}_{it} + \delta_{it}, \tag{11}$$

$$\text{nonsynch}_{it}^{\text{event}} - \text{nonsynch}_{it}^{\text{ave}} = \alpha + \beta \hat{x}_{it} + \gamma \, \text{controls}_{it} + \varepsilon_{it}, \tag{12}$$

where the event window is $[T-21, T-1]$ and the benchmark period is $[T-252, T-64]$.

Interpretation. The differenced measure nets out the part of synchronization due to index or industry arbitrage and isolates the event-related firm-specific component.

3.11 Alternative measure 2: intraperiod timeliness

The paper also implements an intraperiod timeliness (IPT) approach in the style of McNichols (1984) and Alford et al. (1993):

- sort stocks into quintiles by AT proxy,
- construct zero-cost "perfect foresight" hedge portfolios,
- compare how much abnormal return is accumulated before the announcement in low-AT versus high-AT portfolios.

Interpretation. This is a portfolio-level check that asks the same substantive question: do high-AT stocks reveal less firm-specific information before major news?

4 Identification and instruments (what makes the estimates credible)

4.1 What makes the baseline comparison credible

The main ingredients are:

- multiple AT proxies with consistent signs,
- rich lagged controls,
- stock fixed effects,
- a strong first stage using lagged log price,
- and several alternative measures of information acquisition.

Interpretation. The credibility comes from many pieces lining up, not from one single coefficient.

4.2 Why lagged log price is a plausible instrument

The instrument is attractive because:

- it shifts AT incentives through relative tick size and the difficulty of quote maintenance,
- it is measured before the relevant information-acquisition period,
- and it is strongly correlated with every AT proxy, even net of controls.

The paper shows that the correlation between lagged log price and the first principal component of AT proxies is about 0.64 in raw data and about 0.75 net of all controls.

4.3 Threats to identification and how the paper addresses them

The paper discusses several threats:

- **Reverse causality:** AT may withdraw from stocks with informed traders. The IV is meant to isolate ex ante AT incentives instead.
- **Nominal-price management and stock splits:** addressed by controls, sample restrictions, and split-related robustness checks.
- **Liquidity confounds:** addressed with lagged spreads, capitalization, analyst coverage, and institutional ownership.

4.4 Why the timing evidence is especially persuasive

The day-by-day response-ratio exercise is hard to rationalize with a pure contemporaneous omitted-variable story.

- The coefficients become more negative steadily from day -21 to day 0.

- That pattern is exactly what one expects if fewer informed traders are gradually revealing information over the month.

Interpretation. A widening information gap is a dynamic object. It is more naturally interpreted as reduced information production than as a one-shot statistical artifact.

4.5 Main limitations

- The paper does not directly identify the precise mechanism: screening by liquidity providers and order-anticipation strategies are both consistent with the evidence.
- The main setting is earnings announcements, which may understate broader welfare costs if firms wait to make decisions until after disclosures.
- The welfare discussion is suggestive rather than fully structural.
- The price-jump-ratio filter removes many small events, although the appendix argues included and excluded observations look broadly similar.

4.6 Relation to the literature

- Relative to the microstructure literature, the paper shifts attention from ultra-fast incorporation of public information to the slower incentive to acquire information.
- Relative to Grossman–Stiglitz style theories, it provides empirical evidence that information rents can be eroded by trading technology.
- Relative to the literature on information in prices, it contributes a stock-quarter panel measure of relative information acquisition.

5 Tables and figures

5.1 Table 1: summary statistics

Read:

- Table 1 shows substantial cross-sectional dispersion in all four AT proxies.
- The proxies are strongly, but not perfectly, correlated with one another.
- Odd-lot and cancel-to-trade ratios move together, while trade-to-order ratios and average trade size move in the opposite direction.

Interpretation. The proxies seem to capture a common AT component, but they are not mere duplicates; using four of them separately is informative.

5.2 Table 2: baseline OLS regressions

Read:

- All four AT proxies significantly predict less pre-announcement information acquisition.

Table 1
Summary statistics for algorithmic trading proxies and controls

A. Summaries by stock-event observations

AT proxies	Odd lots	Trades/Orders	Cancels/Trades	Trade size
Mean	−2.43	−3.76	3.31	4.71
SD	0.80	0.63	0.58	0.45
10%	−3.59	−4.61	2.62	4.31
Median	−2.26	−3.72	3.25	4.59
90%	−1.58	−3.00	4.09	5.33
N	53,796	53,844	53,844	54,879

Controls	Market cap.	Price	Ret. vol.	Quoted spr.	#Analysts	*IOR*
Mean	20.71	2.90	−3.96	0.50	1.71	0.57
SD	1.89	1.21	0.58	0.84	0.96	0.28
10%	18.28	1.22	−4.64	0.05	0.00	0.00
Median	20.65	3.06	−4.00	0.23	1.79	0.64
90%	23.20	4.27	−3.22	1.15	2.94	0.88
N	54,879	54,879	54,879	54,595	54,879	54,439

B. Pairwise correlations by stock-event observations

AT proxies	Odd lots	Trades/Orders	Cancels/Trades
Trade-to-order ratio	−0.56		
Cancel-to-trade ratio	0.42	−0.83	
Trade size	−0.94	0.52	−0.34

Controls	Market cap.	Price	Ret. vol.	Quoted spr.	#Analysts
Price	0.77				
Ret. vol.	−0.50	−0.56			
Quoted spr.	−0.60	−0.48	0.33		
#Analysts	0.74	0.51	−0.25	−0.53	
IOR	0.42	0.44	−0.24	−0.40	0.43

This table presents descriptive statistics (top panel) and correlations (bottom panel) among key explanatory and control variables. The first table in each panel summarizes algorithmic trading proxies (all in logs). The second table in each panel presents corresponding summaries for dollar market capitalization, share price, and return volatility (all in logs) from the CRSP data. Additional summaries include the median end-of-minute quoted bid-ask spreads in percent from the TAQ NBBO files (one-second version); the number of analysts as the log of the maximum number of reporting analysts for each stock-event pair in Thomson Reuters I/B/E/S; and the institutional ownership ratio (*IOR*) as the fraction of shares held by 13F filing institutions at the end of the preceding calendar quarter. All proxy variables are rolling 21-day mean values from $[T-21, T-1]$, and control variables are lagged an additional 21 days prior to the announcement date ($[T-42, T-22]$).

Table 2
Determinants of announcement price impact

	x = Odd lot ratio			x = Trade-to-order ratio			x = Cancel-to-trade ratio			x = Avg. trade size		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
x	0.0741***	0.0898***	0.144***	−0.0725***	−0.0889***	−0.142***	0.0718***	0.113***	0.144***	−0.152***	−0.150***	−0.263***
	(0.00683)	(0.00937)	(0.0111)	(0.00863)	(0.0103)	(0.0107)	(0.00858)	(0.00988)	(0.0113)	(0.0122)	(0.0176)	(0.0248)
Market cap.	0.00244	−0.0183***	−0.0478***	0.00783***	−0.0280***	−0.0131	0.0135***	−0.0210***	−0.00424	0.000119	−0.0220***	−0.0423***
	(0.00318)	(0.00542)	(0.0121)	(0.00290)	(0.00572)	(0.0118)	(0.00314)	(0.00556)	(0.0119)	(0.00318)	(0.00529)	(0.0124)
Price		−0.0290***	X		−0.00510	X		−0.0105	X		−0.0151*	X
		(0.00867)			(0.00876)			(0.00835)			(0.00857)	
Ret. vol.		−0.0163	−0.000812		−0.0106	0.00528		−0.00193	0.00933		−0.0194	−0.00387
		(0.0124)	(0.0123)		(0.0117)	(0.0122)		(0.0115)	(0.0120)		(0.0120)	(0.0120)
Quoted spr.		−0.0324***	−0.0169		−0.0572***	−0.0363***		−0.0713***	−0.0372***		−0.0222***	−0.00209
		(0.00745)	(0.0123)		(0.00779)	(0.0120)		(0.00871)	(0.0129)		(0.00725)	(0.0115)
#Analysts		0.0521***	0.0326**		0.0546***	0.0361***		0.0564***	0.0334**		0.0505***	0.0305**
		(0.00754)	(0.0136)		(0.00742)	(0.0133)		(0.00746)	(0.0133)		(0.00728)	(0.0132)
IOR		0.123***	0.0239		0.123***	0.0421		0.139***	0.0550		0.106***	0.00781
		(0.0231)	(0.0358)		(0.0222)	(0.0349)		(0.0222)	(0.0347)		(0.0228)	(0.0338)
Constant	0.599***	X	X	0.0403	X	X	−0.0426	X	X	1.179***	X	X
	(0.0729)			(0.0692)			(0.0736)			(0.107)		
Month FEs	No	Yes	Yes	No	Yes	Yes	No	No	Yes	No	No	Yes
Stock FEs	No	No	Yes	No	No	Yes	No	No	Yes	No	No	Yes
R^2	0.0132	0.0337	0.0174	0.00904	0.0330	0.0198	0.00762	0.0352	0.0190	0.0157	0.0331	0.0158
N	24,107	23,814	23,517	24,068	23,800	23,503	24,120	23,842	23,539	24,512	24,201	23,910

$^{*}p<.10$, $^{**}p<.05$, $^{***}p<.01$
This table presents results from a regression of price jump ratios on a set of algorithmic trading proxies:

$$jump_{it}^{(21)} = \alpha + \beta x_{it} + \gamma \times controls_{it} + \epsilon_{it}.$$

For each stock i and quarterly earnings announcement t from January 2012 through September 2016, the price jump ratio ($jump_{it}^{(21)}$) is measured as the ratio of the announcement response divided by the total variation in the pre- and post-announcement period: $CAR_{it}^{(T-1,T+2)}/CAR_{it}^{(T-21,T+2)}$. Cumulative abnormal returns (in logs) are net of Fama and French (1992) three-factor implied returns over the same interval. Observations with cumulative net price impact not exceeding a minimum multiple of prior return volatility (described in the main text) are dropped. Market capitalization, share price, and return volatility are the log of daily averages over $[T-42, T-22]$, and the quoted spread is average of the time-weighted bid-ask spread over the same interval (reported in percent). The number of analysts is the log of the largest number of reporting analysts in the Thomson Reuters I/B/E/S database associated with each stock-quarter announcement. The institutional ownership ratio (*IOR*) is the fraction of shares held by 13F filing institutions at the end of the preceding calendar quarter. Construction of algorithmic trading proxies (x_{it}) derived from SEC MIDAS data are described in the main text. All standard errors are clustered by security and month and are reported in parentheses. The log price is dropped from stock fixed effects regression on account of near-collinearity with market capitalization. Reported R^2 are between-group values in stock fixed effects specifications. The number of observations for the within-stock specifications is slightly smaller on account of dropping securities with only one observation.

- For AT-increasing proxies, the coefficients on the price jump ratio are positive.
- For AT-decreasing proxies, the coefficients are negative.
- The patterns remain after adding controls and stock fixed effects.

Interpretation. Before any IV correction, the cross section and within-stock variation already show a strong negative AT–information-acquisition relationship.

5.3 Table 3 and Figure 1: first-stage relevance of lagged price

Read:

- Table 3 shows that lagged log stock price is strongly correlated with all AT proxies.
- Figure 1 shows the same relation nonparametrically: AT rises monotonically with the decile of lagged log price, both in raw data and after orthogonalizing price with respect to controls.

Interpretation. High-priced stocks appear systematically more algorithmic, consistent with the fine-price-grid mechanism created by the one-cent minimum tick.

5.4 Table 4: IV regressions

Read:

- Table 4 is the core causal table.
- The IV coefficients preserve the signs and magnitudes of the OLS results.
- The first-stage Kleibergen–Paap statistics are extremely large, indicating strong instrument relevance.

Table 3
Correlations with lagged log price instrument

Sample	Odd lots	Trades/orders	Cancels/trades	Trade size	AT PCF_1
Full sample	0.730***	−0.432***	0.251***	−0.766***	0.640***
Regression sample	0.721***	−0.456***	0.293***	−0.752***	0.640***
Net of mkt. cap.	0.787***	−0.587***	0.579***	−0.763***	0.779***
Net of all controls	0.769***	−0.550***	0.560***	−0.750***	0.752***

* $p<.10$, ** $p<.05$, *** $p<.01$
This table reports raw correlations of the lagged log stock price and algorithmic trading proxies for the full sample and for the regression sample, as well as correlations of these measures net of variation spanned by market capitalization and other control variables for each stock i and quarterly earnings announcement t from January 2012 through September 2016. Construction of algorithmic trading proxies (x_{it}) derived from SEC MIDAS data is described in the main text. PCF_1 denotes the first principal component factor of the AT proxies.

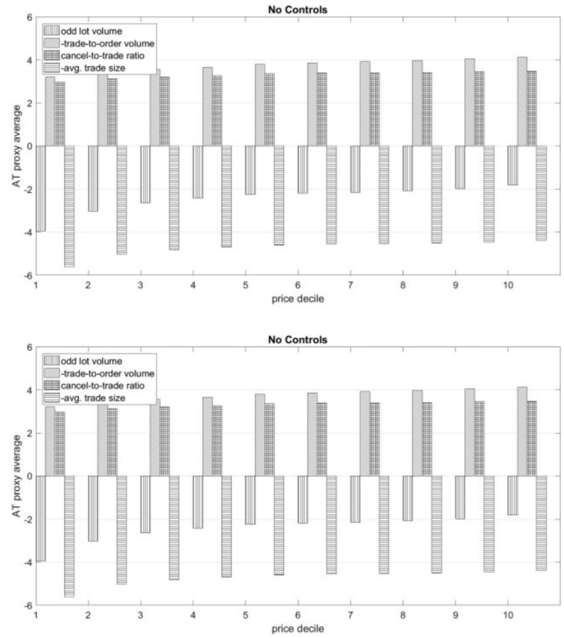

Figure 1
Algorithmic trading proxies by log price decile
This figure presents averages of the four algorithmic trading proxies for each decile of the log price and the component of log price orthogonal to all controls. "No controls" specification (top figure) consists of the algorithmic trading proxy averaged within log price deciles, where deciles are assigned by calendar month. "All controls" specification (bottom figure) averages AT proxies by residual log price decile, where the residual is constructed by regressing the log price on return volatility, quoted spread, number of analysts, institutional ownership ratio, and month fixed effects. AT proxies are normalized to be increasing in AT. Market capitalization, share price, and return volatility are the log of daily averages over $[T-42, T-22]$, and the quoted spread is average of the time-weighted bid-ask spread over the same interval (reported in percent). The number of analysts is the log of the largest number of reporting analysts in the Thomson Reuters I/B/E/S database associated with each stock-quarter announcement. The institutional ownership ratio (IOR) is the fraction of shares held by 13F filing institutions at the end of the preceding calendar quarter. Construction of algorithmic trading proxies derived from SEC MIDAS data is described in the main text.

Table 4
Determinants of announcement price impact with lagged log price instruments

	x = Odd lot ratio		x = Trade-to-order ratio		x = Cancel-to-trade ratio		x = Avg. trade size	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
x	0.0605***	0.0507***	−0.0920***	−0.0778***	0.104***	0.0874***	−0.123***	−0.108***
	(0.00775)	(0.00841)	(0.0127)	(0.0138)	(0.0149)	(0.0158)	(0.0154)	(0.0175)
Market cap.	0.00422	−0.0281***	0.00719**	−0.0296***	0.0146***	−0.0254***	0.00269	−0.0272***
	(0.00320)	(0.00443)	(0.00302)	(0.00439)	(0.00311)	(0.00435)	(0.00321)	(0.00430)
Ret. vol.		−0.0202		−0.0120		−0.00644		−0.0209*
		(0.0127)		(0.0124)		(0.0126)		(0.0123)
Quoted spr.		−0.0395***		−0.0560***		−0.0660***		−0.0292***
		(0.00787)		(0.00792)		(0.00881)		(0.00815)
#Analysts		0.0485***		0.0534***		0.0537***		0.0488***
		(0.00764)		(0.00789)		(0.00775)		(0.00747)
IOR		0.128***		0.123***		0.136***		0.113***
		(0.0227)		(0.0222)		(0.0218)		(0.0227)
Constant	0.529***	X	−0.0193	X	−0.170**	X	0.991***	X
	(0.0747)		(0.0678)		(0.0814)		(0.120)	
Month FEs	No	Yes	No	Yes	No	Yes	No	Yes
Stock FEs	No	No	No	No	No	No	No	No
N	24,107	23,814	24,068	23,800	24,120	23,842	24,512	24,201
K-P rk LM	42.67***	41.92***	42.03***	41.14***	41.44***	41.00***	42.12***	41.44***
K-P rk Wald F	4,226.2	4,554.4	1,493.9	1,189.7	1,001.3	899.2	2,765.0	2,808.9

* $p<.10$, ** $p<.05$, *** $p<.01$
This table reports results from an instrumental variables regression of price jump ratios on a set of algorithmic trading proxies:

$$x_{it} = \zeta + \eta lprice_{it} + \theta \times controls_{it} + \delta_{it},$$

$$jump_{it}^{(21)} = \alpha + \beta \hat{x}_{it} + \gamma \times controls_{it} + \epsilon_{it}.$$

For each stock i and quarterly earnings announcement t from January 2012 through September 2016, the first-stage regression instruments algorithmic trading measures using the log of the average end-of-day stock price from $T-42$ to $T-22$. The second-stage regression for which results are reported relates predicted algorithmic trading proxies to the price jump ratio ($jump_{it}^{(21)}$). The price jump ratio is measured as the ratio of the announcement response divided by the total variation in the pre- and post-announcement period: $CAR_{it}^{(T-1,T+2)}/CAR_{it}^{(T-21,T+2)}$. Cumulative abnormal returns (in logs) are net of Fama and French (1992) three-factor implied returns over the same interval. Observations with cumulative net price impact not exceeding a minimum multiple of prior return volatility (described in the main text) are dropped from both stages. Market capitalization, share price, and return volatility are the log of daily averages over $[T-42, T-22]$, and the quoted spread is average of the time-weighted bid-ask spread over the same interval (reported in percent). The number of analysts is the log of the largest number of reporting analysts in the Thomson Reuters I/B/E/S database associated with each stock-quarter announcement. The institutional ownership ratio (IOR) is the fraction of shares held by 13F filing institutions at the end of the preceding calendar quarter. Construction of algorithmic trading proxies (x_{it}) derived from SEC MIDAS data is described in the main text. All standard errors are clustered by security and month and are reported in parentheses. K-P refers to the Kleibergen and Paap (2006) rk LM and Wald F statistics. In these specifications, a 10% maximal IV size corresponds to a critical value of 16.38.

Main quantitative message.

- A one-standard-deviation increase in AT lowers pre-announcement information acquisition by roughly 9% to 13%.
- The finding is not specific to one AT measure.

5.5 Figure 2 and Table 5: timing of the information gap

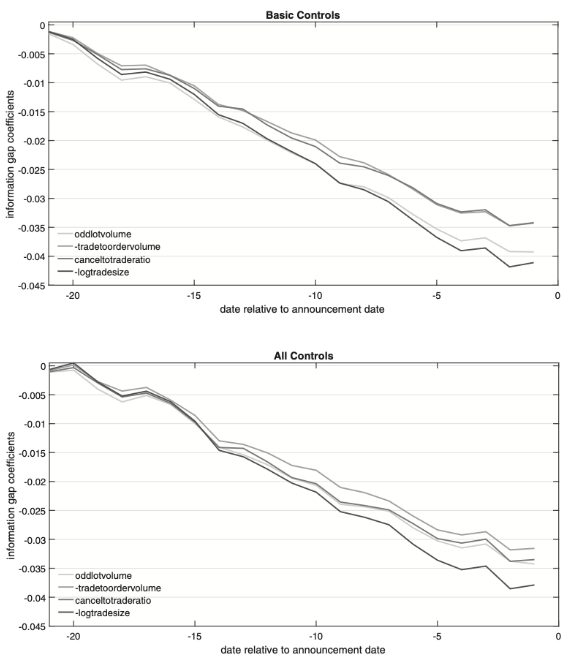

Figure 2
Price response ratios for algorithmic trading proxies
This figure presents coefficients from regressions of price response ratios on the four algorithmic trading proxies using the methodology of Table 5. To facilitate comparison across AT measures, I standardize all coefficients by normalizing the instrumented AT proxy $\hat{x}^{(k)}$ to have a unit standard deviation and to be increasing in AT. For each stock i and quarterly earnings announcement t from January 2012 through September 2016, the price response ratio is measured as the ratio of the cumulative price response through date k prior to the announcement date divided by the total variation over the information incorporation window: $CAR_{it}^{(T-21,T-k)}/CAR_{it}^{(T-21,T+2)}$. Cumulative abnormal returns (in logs) are net of Fama and French (1992) three-factor implied returns over the same interval. Observations with cumulative net price impact not exceeding a minimum multiple of prior return volatility (described in the main text) are dropped. "Basic" specification (top figure) consists of the algorithmic trading proxy and market capitalization. "All" specification (bottom figure) adds return volatility, quoted spread, number of analysts, institutional ownership, and month fixed effects. Market capitalization, share price, and return volatility are the log of daily averages over $[T-42-k, T-22-k]$, and the quoted spread is average of the time-weighted bid-ask spread over the same interval (reported in percent). The number of analysts is the log of the largest number of reporting analysts in the Thomson Reuters I/B/E/S database associated with each stock-quarter announcement. The institutional ownership ratio (IOR) is the fraction of shares held by 13F filing institutions at the end of the preceding calendar quarter. Construction of algorithmic trading measures derived from SEC MIDAS data is described in the main text.

Table 5
Algorithmic trading and the timing of information acquisition

		Δ: 2–4 weeks pre		Δ: 0–2 weeks pre	
		(3)	(4)	(5)	(6)
Controls		Basic	All	Basic	All
Degrees of freedom		10	10	10	10
x = Odd lot ratio	χ^2	53.62***	35.47***	98.61***	54.38***
	p	0.000	0.000	0.000	0.000
x = Trade-to-order ratio	χ^2	45.79***	34.89***	93.80***	56.27***
	p	0.000	0.000	0.000	0.000
x = Cancel-to-trade ratio	χ^2	46.73***	35.56***	91.18***	56.19***
	p	0.000	0.000	0.000	0.000
x = Avg. trade size	χ^2	46.56***	35.11***	97.44***	55.66***
	p	0.000	0.000	0.000	0.000
x = First PCF of AT measures	χ^2	48.78***	34.61***	89.39***	53.79***
	p	0.000	0.000	0.000	0.000
		Δ: 2–4 weeks pre		Δ: 0–2 weeks pre	
		(3)	(4)	(5)	(6)
Controls		Basic	All	Basic	All
Degrees of freedom		10	10	10	10
x = Odd lot ratio	χ^2	51.18***	35.00***	42.01***	22.93**
	p	0.000	0.000	0.000	0.011
x = Trade-to-order ratio	χ^2	44.03***	34.89***	41.34***	24.44***
	p	0.000	0.000	0.000	0.007
x = Cancel-to-trade ratio	χ^2	45.69***	35.96***	41.32***	25.42***
	p	0.000	0.000	0.000	0.005
x = Avg. trade size	χ^2	45.89***	35.47***	50.22***	28.08***
	p	0.000	0.000	0.000	0.002
x = First PCF of AT measures	χ^2	46.30***	34.06***	35.27***	22.20**
	p	0.000	0.000	0.000	0.014

* $p<.10$, **p<.05, *** $p<.01$
This table presents results from IV regressions of price response ratios on a set of algorithmic trading proxies:

$$x_{it}^{(k)} = \zeta + \eta lprice_{it}^{(k)} + \theta \times controls_{it} + \delta_{it},$$

$$responseratio_{it}^{(k,21)} = \alpha + \beta^{(k)} \hat{x}_{it}^{(k)} + \gamma \times controls_{it} + \epsilon_{it}, \ \forall k = 1, \ldots, 21.$$

For each stock i and quarterly earnings announcement t from January 2012 through September 2016, the price response ratio is measured as the ratio of the cumulative price response (first subtable) through date k prior to the announcement date divided by the total variation over the information incorporation window: $CAR_{it}^{(T-21,T-k)}/CAR_{it}^{(T-21,T+2)}$. Second subtable replaces the cumulative price response with the concurrent price response $\left(CAR_{it}^{(T-21,T-k)} - CAR_{it}^{(T-21,T-k-1)}\right)/CAR_{it}^{(T-21,T+2)}$, where I set $CAR_{it}^{(T-21,T-22)}$ equal to zero. Cumulative abnormal returns (in log) are net of Fama and French (1992) three-factor implied returns over the same interval. Observations with cumulative net price impact not exceeding a minimum multiple of prior return volatility (described in the main text) are dropped.

Table entries correspond with cross-equation Wald test statistics and p-values for hypotheses on sets of $\beta^{(\cdot)}$ estimated using two-step GMM. "Basic" specification consists of algorithmic trading proxy and market capitalization. "All" specification adds share price, return volatility, quoted spread, number of analysts, and month fixed effects. Market capitalization, share price, and return volatility are the log of daily averages over $[T-42, T-22]$, and the quoted spread is average of the time-weighted bid-ask spread over the same interval (reported in percent). The number of analysts is the log of the largest number of reporting analysts in the Thomson Reuters I/B/E/S database associated with each stock-quarter announcement. The institutional ownership ratio (IOR) is the fraction of shares held by 13F filing institutions at the end of the preceding calendar quarter. Construction of algorithmic trading proxies ($x_{it}^{(k)}$) derived from SEC MIDAS data is described in the main text, and the first principal component factor of these proxies is denoted by "first PCF." All standard errors are clustered by security and month.

Read:

- Figure 2 plots the information-gap coefficients day by day through the 21 trading days before earnings announcements.
- The coefficients grow more negative as the announcement date approaches.
- Table 5 formally rejects the null of no information gap both in the early subperiod (2–4 weeks before) and in the late subperiod (0–2 weeks before).

Interpretation. The evidence is not that AT only affects prices right before disclosure. It appears to discourage information acquisition throughout the entire month leading into the announcement.

5.6 Table 6 and Table 7: broader events and price nonsynchronicity

Table 6
RavenPack news items

Rank	Group	Type	N	%	Cumulative %
1	Earnings	Earnings	3,903,482	17.7	17.7
2	Products-services	Product-release	1,475,898	6.7	24.4
3	Labor-issues	Executive-appointment	1,235,878	5.6	30.1
4	Products-services	Business-contract	1,184,949	5.4	35.4
5	Revenues	Revenue	876,281	4.0	39.4
6	Analyst-ratings	Analyst-ratings-change	875,978	4.0	43.4
7	Acquisitions-mergers	Scquisition	865,047	3.9	47.3
8	Marketing	Conference	766,278	3.5	50.8
9	Credit-ratings	Credit-rating-change	670,356	3.0	53.8
10	Earnings	Earnings-per-share	595,781	2.7	56.6
11	Investor-relations	Conference-call	577,830	2.6	59.2
12	Insider-trading	Insider-sell	571,758	2.6	61.8
13	Dividends	Dividend	541,002	2.5	64.2
14	Partnerships	Partnership	490,267	2.2	66.5
15	Earnings	Earnings-guidance	480,604	2.2	68.6
16	Insider-trading	Sell-registration	472,851	2.1	70.8
17	Acquisitions-mergers	Unit-acquisition	393,994	1.8	72.6
18	Insider-trading	Insider-buy	370,203	1.7	74.3
19	Price-targets	Price-target	354,128	1.6	75.9
20	Labor-issues	Executive-resignation	302,980	1.4	77.2
21	Assets	Facility	268,681	1.2	78.5
22	Revenues	Revenue-guidance	254,041	1.2	79.6
23	Analyst-ratings	Analyst-ratings-set	251,252	1.1	80.8

This table presents all RavenPack company news categories with shares exceeding 1% of the January 2012–March 2016 RavenPack sample. Technical analysis signals, stock price movements, order imbalance reports, and investor relations items (typically announcements of future information revelation dates) are excluded.

Table 7
Determinants of announcement price impact: Difference of price nonsychronicities

	x = Odd lot ratio		x = Trade-to-order ratio		x = Cancel-to-trade ratio		x = Avg. trade size	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
x	−0.0170***	−0.00736***	0.0286***	0.0115***	−0.0312***	−0.0121***	0.0332***	0.0141***
	(0.00294)	(0.00227)	(0.00492)	(0.00385)	(0.00567)	(0.00434)	(0.00576)	(0.00456)
Market cap.	0.00630***	0.00688***	0.00515***	0.00698***	0.00318	0.00639***	0.00696***	0.00671***
	(0.00192)	(0.00200)	(0.00189)	(0.00198)	(0.00194)	(0.00197)	(0.00194)	(0.00199)
Ret. vol.		−0.0106**		−0.0117**		−0.0124**		−0.0103**
		(0.00481)		(0.00502)		(0.00505)		(0.00481)
Quoted spr.		−0.00309		−0.000796		0.000119		−0.00464
		(0.00362)		(0.00413)		(0.00440)		(0.00352)
#Analysts		−0.00131		−0.00171		−0.00158		−0.00104
		(0.00278)		(0.00276)		(0.00273)		(0.00271)
IOR		−0.00311		−0.00463		−0.00579		−0.00123
		(0.00728)		(0.00754)		(0.00752)		(0.00721)
Constant	−0.239***	X	−0.0674	X	−0.0313	X	−0.368***	X
	(0.0397)		(0.0411)		(0.0448)		(0.0510)	
Month FEs	No	Yes	No	Yes	No	Yes	No	Yes
Stock FEs	No	No	No	No	No	No	No	No
N	53,679	53,004	53,728	53,113	53,727	53,108	54,757	54043
K-P rk LM	43.05***	42.24***	42.54***	41.68***	41.57***	40.90***	42.39***	41.72***
K-P rk Wald F	4,527.7	5,021.6	1,442.0	1,123.1	1,193.1	974.3	3,027.8	3,077.3

* $p < .10$, ** $p < .05$, *** $p < .01$
This table presents results from a regression of price nonsynchronicity differences on a set of algorithmic trading proxies:

$$x_{it} = \zeta + \eta lprice_{it} + \theta \times controls_{it} + \delta_{it},$$

$$nonsynch_{it}^{event} - nonsynch_{it}^{ave} = \alpha + \beta \hat{x}_{it} + \gamma \times controls_{it} + \epsilon_{it}.$$

For each stock i and event date t from January 2012 through September 2016, the average price nonsynchronicity ($nonsynch_{it}^{ave}$) is measured as the average of one minus the R^2 of a regression of returns of stock i on market and sector returns for the year preceding the date of the news event (trading days $T-252$ through $T-1$). The event-period price nonsynchronicity ($nonsynch_{it}^{event}$) is the same quantity measured over the pre-event period of $T-21$ to $T-1$ days. Market capitalization, share price, and return volatility are the log of daily averages over $[T-42, T-22]$, and the quoted spread is average of the time-weighted bid-ask spread over the same interval (reported in percent). The number of analysts is the log of the largest number of reporting analysts in the Thomson Reuters I/B/E/S database associated with the earnings announcement in the same stock-quarter. The institutional ownership ratio (*IOR*) is the fraction of shares held by 13F filing institutions at the end of the preceding calendar quarter. Construction of algorithmic trading proxies (x_{it}) derived from SEC MIDAS data is described in the main text. All standard errors are clustered by security and month and are reported in parentheses. The log price is dropped from stock fixed effects regression on account of near-collinearity with market capitalization. K-P refers to the Kleibergen and Paap (2006) rk LM and Wald F statistics. In these specifications, a 10% maximal IV size corresponds to a critical value of 16.38.

Read:

- Table 6 summarizes RavenPack event categories and shows that earnings are important but not dominant; the news sample also contains many product releases, executive changes, revenue items, analyst updates, and M&A events.
- RavenPack contributes 72,544 additional events.
- Table 7 shows that, after differencing out typical-period synchronization, more AT is associated with less event-period firm-specific information in prices.

Interpretation. The negative effect of AT on information acquisition is not confined to earnings news and is not simply an artifact of index arbitrage.

5.7 Figure 3 and Table 8: intraperiod timeliness

Read:

- Figure 3 compares low-AT and high-AT quintile portfolios. Low-AT portfolios accumulate relatively more abnormal return before the announcement.
- Table 8 rejects equality of cumulative abnormal portfolio returns and equality of IPT measures between high-AT and low-AT portfolios for every AT proxy.
- The differences are large; for example, the Q5–Q1 cumulative abnormal return difference is about −20.36 for odd-lot ratios and −23.40 for average trade size, and the IPT area differences are even larger in magnitude.

Interpretation. The portfolio-level evidence tells the same story as the stock-level evidence: information arrives later in high-AT stocks.

5.8 Appendix Figure A1: sample selection and the jump-ratio filter

Read:

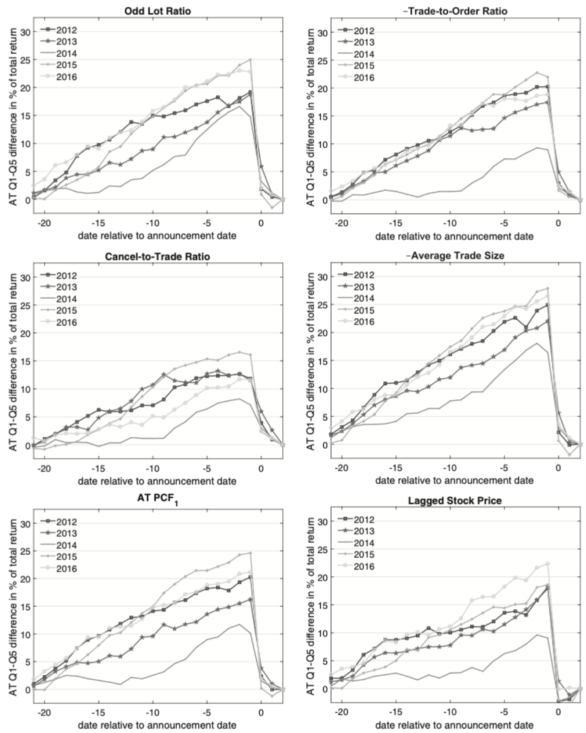

Figure 3
Differences in intraperiod timeliness
This figure presents differences in intraperiod timeliness in the mode of McNichols (1984) and Alford et al. (1993). I sort stocks into quintiles for each algorithmic trading proxy for each calendar year. For securities in quintiles one or five, I construct the "perfect foresight" return on a zero-cost portfolio with scale of one dollar. At date $T-21$ relative to announcement date T, I invest $+1/N_{up}$ dollars in each stock that earns a positive return through $T+2$, and I invest $-1/N_{down}$ dollars in each stock that earns a negative return through $T+2$, where N_{up} and N_{down} are the counts of securities in each group. For each date, I then compute the abnormal portfolio return for each quintile (net of Fama and French 1992 three-factor returns) divided by the total abnormal portfolio return as of $T+2$. Figures plot the difference between the relative returns as of date k preceding the announcement on the low-AT portfolio net of the high-AT portfolio. PCF_1 denotes the first principal component factor of the AT proxies.

Table 8
Intraperiod timeliness tests

A. Test of equality of cumulative abnormal portfolio return as of T

$x=$	Odd lots	$\frac{\text{Trades}}{\text{Orders}}$	$\frac{\text{Cancels}}{\text{Trades}}$	Trade size	AT PCF_1	log price
	(1)	(2)	(3)	(4)	(5)	(6)
Q5–Q1	−20.36***	−16.63***	−11.71***	−23.40***	−18.71***	−16.47***
	(14.19)	(11.38)	(8.85)	(15.06)	(11.62)	(10.84)

B. Test of equality of IPT as of T

$x=$	Odd lots	$\frac{\text{Trades}}{\text{Orders}}$	$\frac{\text{Cancels}}{\text{Trades}}$	Trade size	AT PCF_1	log price
	(1)	(2)	(3)	(4)	(5)	(6)
Q5–Q1	−230.21***	−195.23***	−136.23***	−268.61***	−221.78***	−175.57***
	(11.51)	(10.22)	(7.33)	(13.13)	(10.05)	(8.39)

* $p<.10$, ** $p<.05$, *** $p<.01$

This table presents results from intraperiod timeliness tests in the style of McNichols (1984) and Alford et al. (1993). I sort stocks into quintiles for each algorithmic trading proxy for each calendar quarter. For securities in quintiles one or five, I construct the "perfect foresight" return on a zero-cost portfolio with scale of one dollar. At date $T-21$ relative to announcement date T, I invest $+1/N_{up}$ dollars in each stock that earns a positive return through $T+2$, and I invest $-1/N_{down}$ dollars in each stock that earns a negative return through $T+2$, where N_{up} and N_{down} are the counts of securities in each group. For each date, I then compute the abnormal portfolio return for each quintile (net of Fama and French (1992) three-factor returns) divided by the total abnormal portfolio return as of $T+2$. Panel A tests the hypothesis that the average relative returns (in %) are equal in portfolios 1 and 5 as of date T across calendar quarters. Panel B supplements this t test with an IPT comparison by cumulating the area under each curve and comparing average areas as of date T between portfolios 1 and 5. PCF_1 denotes the first principal component factor of the AT proxies. Standard errors are reported in parentheses.

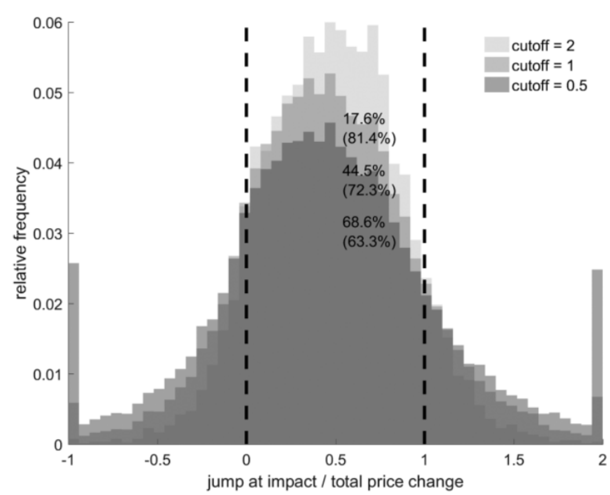

Figure A1
Range of announcement impact price jump ratio
This figure depicts the empirical distribution of earnings announcement price jump ratios for several minimum thresholds for the total price movement over the interval. The price jump ratio is given by $CAR_{it}^{(T-1,T+2)}/CAR_{it}^{(T-21,T+2)}$ for Fama and French (1992) three-factor residualized changes in log price. Stock-quarter observations enter the sample for $\left|CAR_{it}^{(T-21,T+2)}\right| > \sqrt{24}\hat{\sigma}_{it} \times$ cutoff, where $\hat{\sigma}_{it}$ is the daily return volatility from $T-42$ to $T-22$ days before an earnings announcement. For each choice of cutoff in $\{1/2, 1, 2\}$, the top number in the histogram is the proportion of the sample that exceeds the relevance cutoff. The bottom number is the fraction of the conditional ratio of post-announcement date variation to total variation that is bounded in the $[0,1]$ interval. Values exceeding -1 on the left and 2 on the right are winsorized at these respective values for visual clarity.

- Appendix Figure A1 shows the distribution of price jump ratios under different event-size cutoffs.
- The selected cutoff is a compromise between retaining data and keeping the ratio in the economically meaningful range.
- The appendix also argues that included and excluded observations look fairly similar on most observable dimensions.

Interpretation. The results are driven by material events rather than by noisy denominator realizations.

6 Short summary

- **Method.** The paper combines a new event-level measure of pre-disclosure information acquisition with SEC MIDAS order-book data and an instrument based on lagged nominal stock price.
- **Main conceptual result.** AT may improve the speed with which acquired information gets into prices while simultaneously discouraging the acquisition of new information.
- **Main empirical result.** Around earnings announcements, more AT causally reduces pre-announcement information acquisition by roughly 9% to 13% per standard deviation of AT.
- **Timing result.** The information gap emerges well before disclosure and keeps widening through the pre-announcement month.
- **Robustness result.** The finding survives alternative AT proxies, alternative information measures, and broader firm-event samples.
- **Economic lesson.** Evaluating financial technology using only short-run price-efficiency metrics misses a central margin: the incentive to acquire information in the first place.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that algorithmic trading can reduce the amount of information incorporated into stock prices before major firm disclosures.
- This is true even though AT is often associated with faster incorporation of already available public information.
- Therefore, AT can reduce *price informativeness* even when it raises *informational efficiency.*

7.2 Economic intuition

- Information acquisition happens only if informed traders expect to earn rents from their informational advantage.
- If algorithms either screen informed order flow better or anticipate informed trades, those rents shrink.

- Once rents fall, fewer traders invest in acquiring information, so prices become less informative over the economically relevant horizon before important disclosures.

7.3 Takeaway

- The paper's contribution is to show that "faster prices" need not mean "better prices."
- In the specific case of AT, the paper argues that the net informational welfare effect is likely negative, because milliseconds of extra efficiency are much less important for real allocation than a month of missing information before major firm news.
- More broadly, the paper reframes the market-quality debate: one must evaluate technology not only by how well it processes information that already exists, but also by whether it preserves the incentives to create that information.